

Rishaan Kumar

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SUMMARY

Computer Engineering Co-op student who builds and ships full-stack AI applications end-to-end, from database schema and auth layers to LLM orchestration and vector search pipelines. Comfortable driving a feature from architecture to deployment, debugging and implementing fixes under pressure, and merging contributions across subsystems in team environments.

EDUCATION

Computer Engineering, BSc Co-op, University of Alberta

September 2024 – Present

TECHNICAL SKILLS

Backend: Python, FastAPI, Node.js, TypeScript, Express.js, RESTful API design, OAuth2.0, JSON Web Tokens, Role Based Access Control

Databases: PostgreSQL, Supabase (Row Level Security, schema design), ChromaDB, Firebase, AWS

Infrastructure: Docker, Linux, Git/GitHub/GitLab, Bash, Make/CMake, Agile workflows, gdb, Valgrind

Frontend: React, TypeScript, Next.js, HTML, CSS, Figma

Data / AI: R (RStudio), OpenAI APIs, Large Language Model agent orchestration, Retrieval Augmented Generation pipelines, NumPy, Pandas, Matplotlib, OpenCV, PyTorch

EXPERIENCE

Software Engineer | Autonomous Robotic Vehicle Project - U of A | Edmonton, AB

December 2025 - Present

- Built a complete autonomous mission tree for the RoboSub 2026 resupply task from scratch using py_trees and ROS2, implementing a behavior tree architecture with a dynamic object validation loop, blackboard-based state persistence across tree ticks, and a Selector/Retry pattern for continuous environment monitoring.
- Navigated a large undocumented production codebase spanning 20+ ROS2 packages independently, identifying and integrating existing mission components rather than rebuilding them, and correctly applying the team's established blackboard communication pattern across the mission planner.
- Resolved a series of infrastructure blockers including Docker/ROS2 environment conflicts, CMake path errors from a corrupted build cache, missing Python package declarations in setup.py, and a missing shared library blocking Gazebo simulator plugins, restoring full workspace build and simulation capability.

PROJECTS

[Scheduling Agent Platform](#) | Full-Stack | Personal Project

April 2025 - Present

- Created and deployed a production-ready FastAPI backend with REST endpoints, Google OAuth 2.0 authentication, and PostgreSQL (Supabase) with Row Level Security, covering API design, schema design, and access control end-to-end
- Reduced academic schedule planning time by >98% (2 hours to <2 minutes) for myself, and 50+ students via automated schedule generation requiring absolutely minimal user input, and Google Calendar API integration.
- Built and deployed a multi-agent LLM pipeline (controller + overseer architecture) with an Intelligent Document Processing ingestion stage that extracts syllabus deadlines, assessments, and course constraints from unstructured documents.

[Codebase Guide](#) | AI Platform | Hackathon Project

Feb 2026 - Present

- Built a RAG platform reducing codebase onboarding from weeks to 1–2 days; developers query any codebase in natural language with ~3–4s response times via optimized vector search.
- Designed the full ingestion-to-retrieval pipeline: ZIP validation, line-based chunking, OpenAI embedding generation, and ChromaDB vector storage with adaptive retrieval modes.
- Created a hybrid impact analysis engine combining regex word-boundary grep and semantic vector search to compute change-impact risk scores (Low/Medium/High), surfacing file citations and cosine-based confidence metrics.